RACE
Regular Analysis Through Continuous Exercise

## Parabola : RACE-01

### Single Correct Type Questions:

**1.** Length of the latus rectum of the parabola $25[(x-2)^2+(y-3)^2] = (3x-4y+7)^2$ is-

(A) 4 (B) 2 (C) 1/5 (D) 2/5

Find the vertex, axis, latus rectum, and focus of the parabolas:

**MPR208**

### Subjective Type Questions:

**2.** Sketch the graph of following parabola Find

- Length of latus rectum
- co-ordinate of focus, vertex,
- ends of latus rectum
- equation of axis, directrix and tangent at vertex.

(i) $y^2 = x$ **MPR209**

(ii) $y^2 = 10x$ **MPR210**

(iii) $y^2 = -2x$ **MPR211**

(iv) $x^2 = 8y$ **MPR212**

(v) $(y-1)^2 = 4(x-2)$ **MPR213**

(vi) $(x+3)^2 = 8(1-2y)$ **MPR214**

(vii) $(2y+1)^2 = 5-3x$ **MPR215**

(viii) $4y^2 + 12x - 20y + 67 = 0$ **MPR216**

**3.** Find the equation of following parabola:

(i) focus is (1, 1) and directrix is $y + x + 2 = 0$.

**MPR217**

(ii) focus is (0, 1) and vertex (0,2)

**MPR218**

(iii) length of latus rectum is 8, vertex is (1,2) and axis is parallel to x-axis

**MPR219**

(iv) whose latus rectum is 4 units, axis is the line $3x + 4y = 4$ and the tangent at the vertex is the line $4x - 3y + 7 = 0$.

**MPR220**

(v) Vertex (5, 3) and directrix $3x - 4y + 1 = 0$

**MPR221**

**4.** Name the conic represented by the equation $\sqrt{ax}+\sqrt{by}=1$, where $a, b \in R, a, b, > 0$.

**MPR222**

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**5.** Prove that the equation to the parabola, whose vertex and focus are on the axis of $x$ at distances $a$ and $a'$ from the origin respectively, is $y^2 = 4(a' - a)(x - a)$.

**MPR223**

**6.** $PQ$ is a double ordinate of a parabola $y^2 = 4ax$. Find the locus of its point of trisection.

**MPR224**

**7.** A double ordinate of the curve $y^2 = 4px$ is of length $8p$ ; prove that the lines from the vertex to its two ends are at right angles.

**MPR225**

**8.** '$O$' is the vertex of the parabola $y^2 = 4ax$ & $L$ is the upper end of the latus rectum. If $LH$ is drawn perpendicular to $OL$ meeting $OX$ in $H$, prove that the length of the double ordinate through $H$ is $4a\sqrt{5}$.

**MPR226**

**9.** A point $P$ on a parabola $y^2 = 4x$, the foot of the perpendicular from it upon the directrix, and the focus are the vertices of an equilateral triangle, find the area of the equilateral triangle.

**MPR227**

**10.** Prove that for all values of $t$, $x = u\cos\alpha\, t$, $y = u\sin\alpha\, t - \frac{1}{2}gt^2$ represents $a$ parabola and find its' vertex and latus rectum.

**MPR228**

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## ANSWER KEY

**1.** D

**3.** (i) $x^2 + y^2 - 2xy - 8x - 8y = 0$ (ii) $x^2 = -4(y - 2)$

(iii) $(y - 2)^2 = \pm 8(x - 1)^2$ (iv) $\left(\frac{3x+4y-4}{5}\right)^2 = \pm 4\frac{(4x-3y+7)}{5}$

(v) $\left(\frac{4x+3y-29}{5}\right)^2 = \frac{16}{5}\left(\frac{3x-4y-3}{5}\right)$

**4.** Parabola **6.** $9y^2 = 4ax$ **9.** $4\sqrt{3}$

**10.** $\left(\frac{u^2\sin 2\alpha}{2g}, \frac{u^2\sin^2\alpha}{2g}\right)$, $LR = \frac{2u^2\cos^2\alpha}{g}$



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## Parabola : RACE-02

### Single Correct Type Questions:

**1.** Locus of the feet of the perpendiculars drawn from vertex of the parabola $y^2 = 4ax$ upon all such chords of the parabola which subtend a right angle at the vertex is

(A) $x^2 + y^2 - 4ax = 0$ (B) $x^2 + y^2 - 2ax = 0$

(C) $x^2 + y^2 + 2ax = 0$ (D) $x^2 + y^2 + 4ax = 0$

**MPR229**

**2.** Through the focus of the parabola $y^2 = 2px (p > 0)$ a line is drawn which intersects the curve at $A(x_1,y_1)$ and $B(x_2,y_2)$. The ratio $\frac{y_1y_2}{x_1x_2}$ equals-

(A) 2 (B) –1 (C) –4 (D) some function of $p$

**MPR230**

### Subjective Type Questions:

**3.** Find the value of '$a$' for which the point $(a^2 - 1, a)$ lies inside the parabola $y^2 = 8x$.

**MPR231**

**4.** The focal distance of $a$ point on the parabola $(x - 1)^2 = 16(y - 4)$ is 8. Find the co-ordinates.

**MPR232**

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**5.** Show that the focal chord of parabola $y^2 = 4ax$ makes an angle $\alpha$ with x-axis is of length $4a\,\text{cosec}^2\alpha$.

**MPR233**

**6.** If one end of focal chord of parabola

(i) $y^2 = 8x$ is (8, – 8), find other end.

**MPR234**

(ii) $y^2 = -2x$ is (–2, 2), find other end.

**MPR235**

(iii) $x^2 = 4y$ is (–4,4), find other end.

**MPR236**

(iv) $(y - 1)^2 = 4 - 8x$ is (–4,–5), find other end.

**MPR237**

**7.** Find the length of focal chord $AB$ of parabola when

(i) length of latus rectum is 8, '$S$' be the focus and $\frac{AS}{BS} = \frac{3}{4}$

**MPR238**

(ii) Length of latus rectum is 4, perpendicular distance of focal chord from vertex is 3.

**MPR239**

(iii) Length of latus rectum is 2, slope of focal chord is 2 and slope of axis of parabola is –1.

**MPR240**



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**8.** Find the condition that the straight line $ax + by + c = 0$ touches the parabola $y^2 = 4kx$.

**MPR241**

**9.** Find the length of the chord of the parabola $y^2 = 8x$, whose equation is $x + y = 1$.

**MPR242**

**10.** Prove that the locus of the middle points of all chords of the parabola $y^2 = 4ax$ which are drawn through the vertex is the parabola $y^2 = 2ax$.

**MPR243**

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**ANSWER KEY**

| | | | | | |
|---|---|---|---|---|---|
| **1.** | A | **2.** | C | **3.** | $\left(-\infty, -\sqrt{\frac{8}{7}}\right) \cup \left(\sqrt{\frac{8}{7}}, \infty\right)$ |
| **4.** | (–7, 8), (9, 8) | | | | |
| **6.** | (i) (1/2,2) | (ii) (–1/8,–1/2) | (iii) (1,1/4) | (iv) (–7/8,11/3) | |
| **7.** | (i) $\frac{49}{6}$ (ii) $\frac{4}{9}$ (iii) $\frac{20}{9}$ | **8.** | $kb^2 = ac$ | **9.** | $8\sqrt{3}$ |



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## Parabola : RACE-03

### Single Correct Type Questions:

**1.** The straight line joining any point $P$ on the parabola $y^2 = 4ax$ to the vertex and perpendicular from the focus to the tangent at $P$, intersect at $R$, then the equation of the locus of $R$ is

(A) $x^2 + 2y^2 - ax = 0$ (B) $2x^2 + y^2 - 2ax = 0$

(C) $2x^2 + 2y^2 - ay = 0$ (D) $2x^2 + y^2 - 2ay = 0$

**MPR244**

**2.** Through the vertex $O$ of the parabola, $y^2 = 4ax$ two chords $OP$ & $OQ$ are drawn and the circles on $OP$ & $OQ$ as diameters intersect in $R$. If $\theta_1, \theta_2$ & $\phi$ are the angles made with the axis by the tangents at $P$ & $Q$ on the parabola & by $OR$ then the value of, $\cot\theta_1 + \cot\theta_2 =$

(A) $-2\tan\phi$ (B) $-2\tan(\pi - \phi)$ (C) 0 (D) $2\cot\phi$

**MPR245**

### Subjective Type Questions:

**3.** Find the equation of tangent(s) to the parabola

(i) $y^2 = 2x$ whose slope is $\frac{5}{4}$

**MPR246**

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(ii) $x^2 = -2y$ which is parallel to $4x + y + 1 = 0$

**MPR247**

(iii) $y^2 = 12x$, which passes through the point (2,5).

**MPR248**

(iv) $y^2 = 8x$ and making an angle of 45° with the line $y = 3x + 5$.

**MPR249**

**4.** Find the point of contact when

(i) Equation of parabola is $x^2 = -8y$ and slope of tangent is 3.

**MPR250**

(ii) Equation of parabola is $y^2 = 4(1 - x)$ and slope of tangent is $-1$.

**MPR251**

**5.** Find the point of intersection of tangents drawn at

(i) the points (2,–4) and $\left(\frac{1}{2}, -2\right)$ of the parabola $y^2 = 8x$

**MPR252**

(ii) the points (–2,2) and (–8,–4) of parabola $y^2 = -2x$.

**MPR253**

**6.** Prove that the locus of the point of intersection of tangents to the parabola $y^2 = 4ax$ which meet at an angle $\theta$ is $(x + a)^2 \tan^2\theta = y^2 - 4ax$.

**MPR254**



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**7.** Prove that the straight line $\ell x + my + n = 0$ touches the parabola $y^2 = 4ax$ if $\ell n = am^2$.

**MPR255**

**8.** Find the equation of common tangent(s) of
(i) Circle $x^2 + y^2 = 4ax$ and parabola $y^2 = 4ax$.

**MPR256**

(ii) Parabola $x^2 = -8y$ and $y^2 = -x$.

**MPR257**

**9.** Tangents are drawn to the parabola $y^2 = 12x$ at the points $A, B$ and $C$ such that the three tangents form a triangle $PQR$. If $\theta_1, \theta_2$ and $\theta_3$ be the inclinations of these tangents with the axis of $x$ such that their cotangents form an arithmetical progression (in the same order) with common difference 2. Find the area of the triangle $PQR$.

**MPR258**

**10.** Two tangents to the parabola $y^2 = 8x$ meet the tangent at its vertex in the points $P$ & $Q$. If $PQ = 4$ units, prove that the locus of the point of the intersection of the two tangents is $y^2 = 8(x + 2)$.

**MPR259**

**11.** Let a variable point $A$ be lying on the directrix of parabola $y^2 = 4x$. Tangents $AB$ & $AC$ are drawn to the curve where $B$ & $C$ are points of contact of tangent. If the locus of centroid of $\Delta ABC$ is a conic then find the length of its latus rectum.

**MPR260**

**12.** Find the locus of the point $P$ from which tangents are drawn to the parabola $y^2 = 4ax$ having slopes $m_1$ and $m_2$ such that -
(i) $m_1^2 + m_2^2 = 10$

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**MPR261**

(ii) $\theta_1 - \theta_2 = 30°$ (constant)

**MPR262**

where $\theta_1$ and $\theta_2$ are the inclinations of the tangents from positive x-axis.

**ANSWER KEY**

| | | | |
|---|---|---|---|
| **1.** | B | **2.** | A |
| **3.** | (i) $25x - 20y + 8 = 0$ | | (ii) $4x + y - 8 = 0$ |
| | (iii) $x - y + 3 = 0, 3x - 2y + 4 = 0$ | | (iv) $2x + y + 1 = 0, x - 2y + 8 = 0$ |
| **4.** | (i) $(-12,-18)$ (ii) $(0,2)$ | **5.** | (i) $(1,-3)$ (ii) $(4,-1)$ |
| **8.** | (i) $x = 0$ (ii) $x + 2y - 1 = 0$ | **9.** | 72 |
| **11.** | 3 | **12.** | (i) $10x^2 - y^2 + 2ax = 0$ (ii) $3(y^2 - 4ax) = (x + a)^2$ |



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## Parabola : RACE-04

### Single Correct Type Questions:

**1.** If three distinct and real normals can be drawn to $y^2 = 8x$ from the point $(a, 0)$, then -

(A) $a > 2$ (B) $a \in (2, 4)$

(C) $a > 4$ (D) none of these

**MPR263**

**2.** If two normals to a parabola $y^2 = 4ax$ intersect at right angles then the chord joining their feet pass through a fixed point whose co-ordinates are:

(A) $(-2a, 0)$ (B) $(a, 0)$

(C) $(2a, 0)$ (D) none

**MPR264**

**3.** If the normal to a parabola $y^2 = 4ax$ at $P$ meets the curve again in $Q$ and if $PQ$ and the normal at $Q$ makes angles $\alpha$ and $\beta$ respectively with the x-axis then $\tan\alpha\,(\tan\alpha + \tan\beta)$ has the value equal to

(A) 0 (B) –2

(C) $-\frac{1}{2}$ (D) –1

**MPR265**

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**4.** If a normal to a parabola $y^2 = 4ax$ makes an angle $\phi$ with its axis, then it will cut the curve again at an angle

(A) $\tan^{-1}(2\tan\phi)$ (B) $\tan^{-1}\left(\frac{1}{2}\tan\phi\right)$

(C) $\cot^{-1}\left(\frac{1}{2}\tan\phi\right)$ (D) none

**MPR266**

### Subjective Type Questions:

**5.** Find the number of distinct normal that can be drawn from (–2, 1) to the parabola $y^2 - 4x - 2y - 3 = 0$.

**MPR267**

**6.** If $2x + y + k = 0$ is a normal to the parabola $y^2 = -16x$, then find the value of $k$.

**MPR268**

**7.** Find the equation of normal(s) to the parabola

(i) $y^2 = -2x + 1$ at the point (–4,–3)

**MPR269**

(ii) $y^2 = -2x$ and slope of normal is 3.

**MPR270**

(iii) $(y - 4)^2 = 16x + 16$ and normal passes through (14,7)

**MPR271**



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**8.** Three normals are drawn from the point (7, 14) to the parabola $x^2 - 8x - 16y = 0$. Find the co-ordinates of the feet of the normals.

**MPR272**

**9.** Write down the equations to the normal at the ends of the latus rectum of the parabola $y^2 = 4a(x - a)$

**MPR273**

**10.** Find the point of the parabola $y^2 = 4ax$ at which the normal is inclined at 30° to the axis.

**MPR274**

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**ANSWER KEY**

| | | | | | |
|---|---|---|---|---|---|
| **1.** | C | **2.** | B | **3.** | B |
| **4.** | B | 5. | 1 | **6.** | 48 |
| **7.** | (i) $3x + y + 15 = 0$ | (ii) $6x - 2y + 33 = 0$ | | (iii) $x = 2y, 3x + 2y = 56, x - y - 7 = 0$ | |
| **8.** | (0, 0), (–4, 3) and (16, 8) | **9.** | $(x + y = 4a, x - y = 4a)$ | **10.** | $\left(\frac{a}{3}, -\frac{2\sqrt{3}}{3}a\right)$ |



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## Parabola : RACE-05

### Single Correct Type Questions:

**1.** Normal to the parabola $y^2 = 8x$ at the point $P(2, 4)$ meets the parabola again at the point $Q$. If $C$ is the centre of the circle described on $PQ$ as diameter then the coordinates of the image of the point $C$ in the line $y = x$ are

(A) $(-4, 10)$ (B) $(-3, 8)$

(C) $(4, -10)$ (D) $(-3, 10)$

**MPR275**

**2.** $PQ$ is a normal chord of the parabola $y^2 = 4ax$ at $P$, $A$ being the vertex of the parabola. Through $P$ a line is drawn parallel to $AQ$ meeting the x-axis in $R$. Then the length of $AR$ is -

(A) equal to the length of the latus rectum.

(B) equal to the focal distance of the point $P$.

(C) equal to twice the focal distance of the point $P$.

(D) equal to the distance of the point $P$ from the directrix.

**MPR276**

**3.** Normals are drawn at points $A$, $B$, and $C$ on the parabola $y^2 = 4x$ which intersect at $P(h, k)$. The locus of the point $P$ if the slope of the line joining the feet of two of them is 2, is

(A) $x + y = 1$ (B) $x - y = 3$

(C) $y^2 = 2(x - 1)$ (D) $y^2 = 2\left(x - \frac{1}{2}\right)$

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**MPR277**

**4.** Tangents are drawn from the point $(-1, 2)$ on the parabola $y^2 = 4x$. The length, these tangents will intercept on the line $x = 2$ is :

(A) 6 (B) $6\sqrt{2}$ (C) $2\sqrt{6}$ (D) none of these

**MPR278**

### Subjective Type Questions:

**5.** Three normals to $y^2 = 4x$ pass through the point $(15, 12)$. Show that if one of the normals is given by $y = x - 3$ & find the equations of the others.

**MPR279**

**6.** Prove that, the normal to $y^2 = 12x$ at $(3, 6)$ meets the parabola again in $(27, -18)$ & find the equation of circle on this normal chord as diameter.

**MPR280**

**7.** From the point $P(h, k)$ three normals are drawn to the parabola $x^2 = 8y$ and $m_1$, $m_2$ and $m_3$ are the slopes of three normals

(a) Find the algebraic sum of the slopes of these three normals.

**MPR281**

(b) If two of the three normals are at right angles then the locus of point $P$ is a conic, find the latus rectum of conic.

**MPR282**

(c) If the two normals from $P$ are such that they make complementary angles with the axis then the locus of point P is a conic, find a directrix of conic.

**MPR283**



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**8.** $PC$ is the normal at $P$ to the parabola $y^2 = 4ax$, $C$ being on the axis. $CP$ is produced outwards to $Q$ so that $PQ = CP$; show that the locus of $Q$ is a parabola.

**MPR284**

**9.** Let $P(a, b)$ and $Q(c, d)$ are the two points on the parabola $y^2 = 8x$ such that the normals at them meet in (18,12). Find the product $(abcd)$.

**MPR285**

**10.** Find combined equation of pair of tangents drawn to parabola.

(i) $y^2 = 8x$ from (–3,1)

**MPR286**

(ii) $(x - 1)^2 = 4(y - 2)$ from (2,0)

**MPR287**

**11.** If the equation $\lambda^2 x + \lambda y - \lambda^2 + 2\lambda + 7 = 0$ represents a family of lines, where '$\lambda$' is parameter, then find the equation of the curve to which these lines will always be tangents.

**MPR288**

**12.** Find the angle between the tangents drawn from the origin to the parabola, $y^2 = 4a(x - a)$.

**MPR289**

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**ANSWER KEY**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **1.** | A | **2.** | C | **3.** | B | **4.** | B |
| **5.** | $y = -4x + 72, y = 3x - 33$ | | | **6.** | $x^2 + y^2 - 30x + 12y - 27 = 0$ | | |
| **7.** | (a) 0; (b) 2; (c) $y = \frac{3}{2}$ | | | **9.** | 512 | | |
| **10.** | (i) $[2x^2 - 3y^2 - xy + 13x + 3y + 18 = 0]$ | | | | (ii) $[2x^2 - y^2 + xy - 8x - 2y + 8 = 0]$ | | |
| **11.** | $(y + 2)^2 = 28(x - 1)$ | | | **12.** | $(\pi/2)$ | | |



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## Parabola : RACE-06

### Single Correct Questions

**1.** $TP$ & $TQ$ are tangents to the parabola, $y^2 = 4ax$ at $P$ & $Q$. If the chord $PQ$ passes through the fixed point $(-a, b)$ then the locus of $T$ is :

(A) $ay = 2b\,(x - b)$ (B) $bx = 2a\,(y - a)$

(C) $by = 2a\,(x - a)$ (D) $ax = 2b\,(y - b)$

**MPR293**

### Subjective Type Questions:

**2.** Find the equation of the chord of contacts of tangents drawn from a point (2, 1) to the parabola $x^2 = 2y$.

**MPR290**

**3.** Find the co-ordinates of the middle point of the chord of the parabola $y^2 = 16x$, the equation of which is $2x - 3y + 8 = 0$

**MPR291**

**4.** Prove that the length of the chord joining the points of contact of tangents drawn from the point $(x_1, y_1)$ is $\frac{\sqrt{y_1^2 + 4a^2}\sqrt{y_1^2 - 4ax_1}}{a}$

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**MPR292**

**5.** Through the vertex $O$ of a parabola $y^2 = 4x$, chords $OP$ & $OQ$ are drawn at right angles to one another. Show that for all positions of $P$, $PQ$ cuts the axis of the parabola at a fixed point. Also find the locus of the middle point of $PQ$.

**MPR294**

**6.** From the point $(-1, 2)$ tangent lines are drawn to the parabola $y^2 = 4x$. Find the equation of the chord of contact. Also find the area of the triangle formed by the chord of contact & the tangents.

**MPR295**

**7.** $PQ$, a variable chord of the parabola $y^2 = 4x$ subtends a right angle at the vertex. The tangents at $P$ and $Q$ meet at $T$ and the normals at those points meet at $N$. If the locus of the mid point of $TN$ is a parabola, then find its latus rectum.

**MPR296**

**8.** Prove that the normal chord at the point whose ordinate is equal to its abscissa subtends a right angle at the focus.

**MPR297**

**9.** Prove that the locus of the middle points of all tangents drawn from points on the directrix to the parabola is $y^2\,(2x + a) = a(3x + a)^2$.

**MPR298**



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**10.** A variable circle passes through the point $A(2,1)$ and touches the x-axis. Locus of the other end of the diameter through $A$ is a parabola.

(a) Find the length of the latus rectum of the parabola.

**MPR299**

(b) Find the co-ordinates of the foot of the directrix of the parabola.

**MPR300**

(c) The two tangents and two normals at the extremities of the latus rectum of the parabola constitutes a quadrilateral. Find area of quadrilateral.

**MPR301**

**11.** Let $L_1 : x + y = 0$ and $L_2 : x - y = 0$ are tangent to a parabola whose focus is $S(1, 2)$. If the length of latus-rectum of the parabola can be expressed as $\frac{m}{\sqrt{n}}$ (where $m$ and $n$ are coprime) then find the value of $(m + n)$.

**MPR302**

**12.** Tangent drawn at point $P(1,3)$ of a parabola intersects its tangent at vertex at $A(-1,5)$. If $B(-5,5)$ is a point on $SP$ where $S$ is focus of the parabola then find

(i) focus

**MPR303**

(ii) equation of directrix.

**MPR304**

(iii) co-ordinate of vertex.

**MPR305**

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**ANSWER KEY**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **1.** | C | **2.** | $2x = y + 1$ | **3.** | 14, 12 | **5.** | $y^2 = 2x - 8$ |
| **6.** | $y = x - 1, 8\sqrt{2}$ | **7.** | $\frac{25}{2}$ | | | | |
| **10.** | (a) 4, (b) (2,–1), (c) 8 sq. units | | | **11.** | 11 | | |
| **12.** | (i) (–2, 4) (ii) $x - 3y + 18 = 0$ (iii) $\left(-\frac{11}{5}, \frac{23}{5}\right)$ | | | | | | |



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